

Auteur: Done Goker
Studierichting: Informatica
Oefeningenreeks 4A nr. 27.17

$$\text{Opgave: } \int \frac{x^4 + x^3 + 3x^2 + x + 1}{x^3 + 1} dx$$

$$= \int (x + 1) dx + \int \frac{3x^2}{x^3 + 1} dx$$

$$t = x^3 + 1$$

$$\frac{dt}{3x^2} = dx$$

$$= \frac{x^2}{2} + x + \int \frac{1}{t} dt$$

$$= \frac{x^2}{2} + x + \ln|x^3 + 1| + C$$

References